CardScript

Iteration 1
CIS 320-01
Team: Card Script
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Pet4Life Project
Vision (Small Project)

Version <2.0>

Revision History

Date	Version	Description	Author
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Vision (Small Project)

1. Introduction

This document will summarize and explain the requirements necessary for the renovation of Pet4Life's website and operations. This document will provide focus on the features needed by the Pet4Life organization and the underserved community in Louisville.

1.1 References

- Internally, the Business Process Analysis Summary – Pets4Life Louisville (January 21, 2025 – Present)
- Assignment 1 Group Excel – Business Process Analysis (January 21, 2025) compiled quantitative data from the research, highlighting the effectiveness of various business processes across these organizations.
- Information about Pets4Life – Internal Document (January 21, 2025 – Present) consolidated research findings, stakeholder input, and proposed solutions, serving as a comprehensive reference for developing strategic website enhancements.
- Pets4Life Louisville Official Website (https://pets4lifelou.org), accessed on January 21, 2025.
- WordPress vs. Other CMS: How It Stacks Up (https://www.namecheap.com/guru-guides/wordpress-vs-other-content-management-systems/), an article from Namecheap accessed on January 21, 2025.
- WordPress Payments: Best Payment Methods for Your Business (https://www.bigcommerce.com/blog/wordpress-payments/)

2. Position

2.1 Problem Statement

The problem of	Managing pet wellness services and resources through manual processes and a limited website infrastructure
affects	- Pet owners in underserved communities - Pets4Life administrative team and Board of Directors - Veterinary staff and volunteers - Partner organizations and donors
the impact of which is	- Inefficient service delivery and resource distribution - Limited reach to target communities in 8 Louisville zip codes - Time-consuming manual payment and appointment tracking - Difficulty coordinating pop-up vet clinics - Reduced ability to effectively communicate services and resources
a successful solution would be	- Streamlines appointment scheduling and payment processing - Centralizes resource management and distribution tracking - Provides easy access to service information and resources - Enables efficient volunteer coordination - Automates administrative tasks and reporting - Enhances community outreach capabilities

2.2 Product Position Statement

For	Pet owners in underserved communities of Greater Metro Louisville
Who	Need access to affordable pet wellness resources and veterinary services but face barriers due to location and resources
The (product name)	The Pets4Life Website is a comprehensive digital platform
That	Streamlines access to pet wellness services, resources, and information while improving operational efficiency
Unlike	The current manual processes and basic website
Our product	Will provide an integrated platform for service delivery, resource management, and community engagement that better serves our mission

3. Stakeholder and User Descriptions

3.1 Stakeholder Summary

Name	Description	Responsibilities
Board of Directors:	Organization leadership team including President: Dr. Lisa Keehner, Treasurer: Connie Hunt, Secretary: Cindy Reinhart, and Directors: Rich Smith and Karen Mujica	- Ensures website aligns with organizational mission
Executive Director	Karen Mujica, manages day-to-day operations - Coordinates	- Oversees website implementation

	between stakeholders - Ensures operational requirements are met	
Veterinary Staff	Service providers at pop-up clinics - Provides input on appointment system - Validates service workflows	- Defines clinical service requirements
Administrative Staff	Support team managing operations - Processes payments and appointments - Coordinates volunteer activities	- Manages content and resources

3.2 User Summary

Name	Description	Responsibilities	Stakeholder
Pet Owners	Community members seeking pet wellness services - Access resources - Submit service requests	- Schedule appointments	Executive Director
Clinic Staff	Veterinary service providers	- Manage appointments	Board of Directors

	- Update availability - Record services		
Administrators	Website and system managers - Process payments - Generate reports	- Update content	Executive Director
Volunteers	Support service delivery - Track activities - Access resources	- Sign up for events	Volunteer Coordinator

3.3 User Environment

The Pets4Life Louisville platform must function effectively within a diverse and dynamic environment that accommodates a wide range of users with varying levels of technical proficiency. Given the organization's focus on underserved communities, accessibility is a critical factor. Many users may have limited access to advanced technology, relying on older mobile devices or shared computers to interact with the platform. As a result, the website must be compatible with multiple browsers and optimized for both desktop and mobile use to ensure seamless navigation across different screen sizes.

Internet connectivity in service areas is another key consideration. Many of the regions served by Pets4Life Louisville experience inconsistent or slow internet access, which necessitates a lightweight and responsive website design that minimizes loading times while maintaining full functionality. Additionally, the platform must support multiple concurrent users, particularly during high-demand

periods such as pop-up veterinary clinics and donation drives. These events often attract a significant number of visitors looking to access resources, schedule appointments, or make contributions, requiring the system to efficiently handle traffic spikes without compromising performance.

Furthermore, the platform must integrate with payment processing systems to facilitate donations and service payments, ensuring a secure and straightforward transaction experience for users. For volunteers and staff working in the field, mobile accessibility is crucial, allowing them to manage scheduling, access records, and communicate with pet owners while on-site. The system will be operational during both regular business hours and during special community events, meaning it must remain highly functional and available outside of traditional office settings. By addressing these factors, Pets4Life Louisville can create a digital infrastructure that effectively supports its mission and maximizes engagement with its community.

3.4 Summary of Key Stakeholder or User Needs

Need	Priority	Concerns	Current Solution	Proposed Solutions
Appointment Management	High	Manual tracking and coordination	Calendly with manual data transfer	Plan Hero
Payment Processing	High	Multiple systems, manual reconciliation	Various payment methods with manual tracking	Stripe
Volunteer Management	High	Difficult coordination	SignUpGenius	Plan Hero

Customer Service	High	No way of tracking customer questions	Manually answering questions through many different channels	Tidio Customer Support Help Desk
Communication System	Medium	Scattered channels	Multiple communication methods	MailPoet plugin
Customer Service	High	No on-site automatic customer service.	Manually detailing instructions through many different channels	Tidio Flow, Knowledge-based chatbot

3.5 Alternatives and Competition

An analysis of twelve comparable websites revealed key competitors and alternative approaches in the field of animal welfare and pet support services. Among the most notable organizations are the Shamrock Pet Foundation, which leads with a feature score of 6 out of 7, followed by the Animal Protection Association with a score of 5 out of 7. Other prominent competitors include the Humane Society of Sarasota County, ASPCA, and the Animal Care Society, each scoring 4 out of 7. These organizations set industry benchmarks for effective digital engagement, incorporating a wide range of features that enhance user experience and support operational efficiency.

The competitive advantage of these leading organizations lies in their comprehensive service information, well integrated donation systems, structured volunteer management capabilities, and seamless event coordination features. Many of these websites also provide robust contact management, ensuring that pet owners and donors can easily connect with their respective organizations. These functionalities contribute to a strong online presence, fostering trust and encouraging community involvement.

Despite the competitive landscape, Pets4Life Louisville differentiates itself through its unique focus on underserved communities and its innovative pop-up clinic model.

Unlike traditional organizations with fixed locations, Pets4Life Louisville brings veterinary services directly to areas where pet care resources are scarce, addressing a critical gap in the market. Its targeted geographic service area ensures that aid is concentrated where it is most needed, maximizing impact. Additionally, the organization integrates resource distribution into its outreach efforts, providing not just medical care but also essential pet supplies, food, and educational materials to pet owners facing financial hardship. This community-centric approach positions Pets4Life Louisville as a mission-driven leader in the field, emphasizing accessibility, inclusivity, and direct support for pet families in need.

4. Product Overview

4.1 Product Perspective

Our product is the renovation of Pet4Life's existing website on the WordPress platform. We envision a combination of WordPress plugins that will integrate into the website to create a cohesive system to facilitate Pet4Life's business processes. This new system will provide a fluid execution of processes for the users in the Louisville community and Pet4Life as an organization.

4.2 Assumptions and Dependencies

Pet4Life's budget is a factor that will shape how we renovate their operations. Some plugins needed to meet the requirements of the To-Be product may have a paywall for their use and if we deem the paywall too high for our client, we may have to change which plugins we use. This could further affect the total features that a plugin offers to facilitate any of Pet4Life's operations.

5. Product Features

Navigation Bar

A new navigation bar for the website to improve the experience of exploring the website and increase the flow of information about the organization and the services they provide to the community. Also provides a form of SEO (Search Engine Operation) for the website by improving the structure of the website and increasing its discoverability on search engines.

Volunteer Sign-up

A volunteer sign-up form sheet located directly on the website for those who want to volunteer for Pet4Life events or to help facilitate their services. Users will be able to visit the website and submit a request for volunteer that the Pet4Life organization can view.

Ticketing System

Customer request ticketing system that is available directly on the website for users to use and submit requests or ask questions. Will provide a functional user service platform to fulfill requests and answer questions. The platform will be located on the site and will provide management tools to direct the workflow among staff and to organize tickets to increase efficiency. Will integrate with the website chatbot and all ticket information will be able to be transferred to Pet4Life's existing AirTable database.

Payment System

Payment system that will process transactions and track the details of the transaction such as the cardholder and their pet. Will integrate with the forms plugins that will be used on the website. Transaction data can be transferred to database.

Chatbot

Chatbot accessible on the website that can provide details and instructions based on the user's needs. Users will be able to ask questions to the bot and the bot will provide a response based on the question. This bot will be able to fulfill the requests of users and direct elsewhere if it cannot. This chatbot integrates into the ticketing system by creating a ticket on behalf of the user. This chatbot will also be able to answer Emails that Pet4Life receives from the community.

6. Other Product Requirements

Pet4Life's website is hosted on the WordPress platform, meaning that the scope of our renovations is technically limited to the capabilities that WordPress and its community have developed. There is a myriad of factors that may play into the technical performance of Pet4Life's website. For example, the number of plugins we use on their website may slow-down load times and they may increase the chance of operation down time. Despite the need for WordPress's maximum performance, WordPress platform's performance is not in our control, and we have no way of influencing it.

All the plugins we will use for their website are hosted externally by other organizations. The facilitations of Pet4Life's operations will depend on these plugins. If these plugins undergo down time of any kind, this will halt the operations of Pet4Life. Great security protocols and procedures are needed from WordPress and the organizations that host the plugins to ensure the security of Pet4Life's data as well.

7. Feasibility Analysis

7.1 Narrative

Problem Statement:

Pets4Life Louisville, established in 2015, is committed to improving pet retention and the overall health and welfare of companion animals in Greater Metro Louisville's underserved communities. The organization faces operational challenges due to outdated manual processes, fragmented information systems, and limited resources. These inefficiencies hinder Pets4Life's ability to provide timely and effective services, affecting pet welfare and retention. Key operational issues include inefficient scheduling systems, inadequate volunteer coordination, and ineffective donor engagement strategies. As the organization strives to transition from an all-volunteer model to a sustainable, staff-supported nonprofit, a robust and integrated information system becomes crucial.

Business Case:

The implementation of an integrated information system at Pets4Life Louisville is poised to transform the organization by addressing its current operational inefficiencies. This system will significantly enhance service efficiency by automating appointment scheduling and resource management, potentially reducing booking times by up to 50%. This will enable the organization to extend its reach more effectively within the 40210 ZIP code, potentially increasing the number of pets served by up to 30% annually. Additionally, the new system will bolster financial stability through improved fundraising efforts facilitated by an efficient donor management system, expected to increase donations by 20-25%. Moreover, a user-friendly interface will be developed to enhance interactions with volunteers, donors, and pet owners, thus increasing their engagement and satisfaction levels.

7.2 Feasibility Analysis

Technical Feasibility:

"The proposed information system is meticulously designed to integrate seamlessly with existing organizational tools such as Airtable and WordPress, enhancing these platforms with tailored functionalities to meet Pets4Life's specific needs. The system will support a robust customer service hub equipped with Tidio Flow Customer Service to handle routine inquiries and Tidio’s ticketing system for more complex issues, significantly improving response times and client satisfaction. Additionally, the system will include a clinic management module utilizing Airtable databases to streamline pet registration and medical record management. This module will ensure accurate tracking and easy access to health data and vaccination records. Security

measures will be a top priority, with robust data encryption and strict adherence to privacy regulations to protect sensitive information. The system’s architecture is also designed to be scalable, facilitating future enhancements and integrations, such as Tidio for user service requests and MailPoet for communication management, without disrupting existing operations."

Economic Feasibility:

Economically, the development and implementation of this new system represent a significant investment but promise a substantial return on investment (ROI). By integrating PlanHero for automated scheduling and volunteer management and Tidio for streamlined user service requests, Pets4Life will significantly reduce labor costs associated with manual data entry and administrative tasks. This automation will not only decrease administrative errors but also free up resources to be. redirected towards direct service provision, thereby enhancing the overall effectiveness of the organization’s programs. Additionally, the system’s enhanced donor engagement capabilities, facilitated through targeted functionalities in MailPoet for communication and Tidio Flow Customer Service for improved donor interactions, are expected to increase the frequency and amount of donations. Together, these improvements are projected to achieve a positive ROI within one year of implementation, affirming the financial viability of this initiative.

Organizational Feasibility:

From an organizational perspective, the new system aligns with Pets4Life Louisville's mission and operational goals, marking a crucial advancement in the organization's development. A targeted training program will ensure all staff and volunteers are proficient with the new system, complemented by ongoing support for any emerging issues. Features like the Tidio chatbot, Tidio help desk and MailPoet will enhance communication and collaboration across the organization, leading to increased operational efficiency and workplace morale.

Also, tools like Tidio’s chatbot and help desk will improve response times and free up staff to focus on direct service provision, reinforcing the organization's commitment to its community-focused objectives.

In conclusion, this strategic information system is designed to seamlessly integrate into current operations, enhance service delivery, and support Pets4Life Louisville’s growth, making it a vital asset in advancing the organization’s mission.

8. Initial Architecture Considerations

The Pets4Life Louisville system is being developed as a fully integrated digital platform to improve clinic management, volunteer coordination, donor engagement, and customer service. Built on **WordPress**, the platform incorporates a variety of plugins and integrations to streamline administrative tasks, enhance user experience, and optimize workflow. By consolidating these functions into a single, cohesive system, Pets4Life Louisville will be able to improve efficiency, reduce operational bottlenecks, and provide better services to the community.

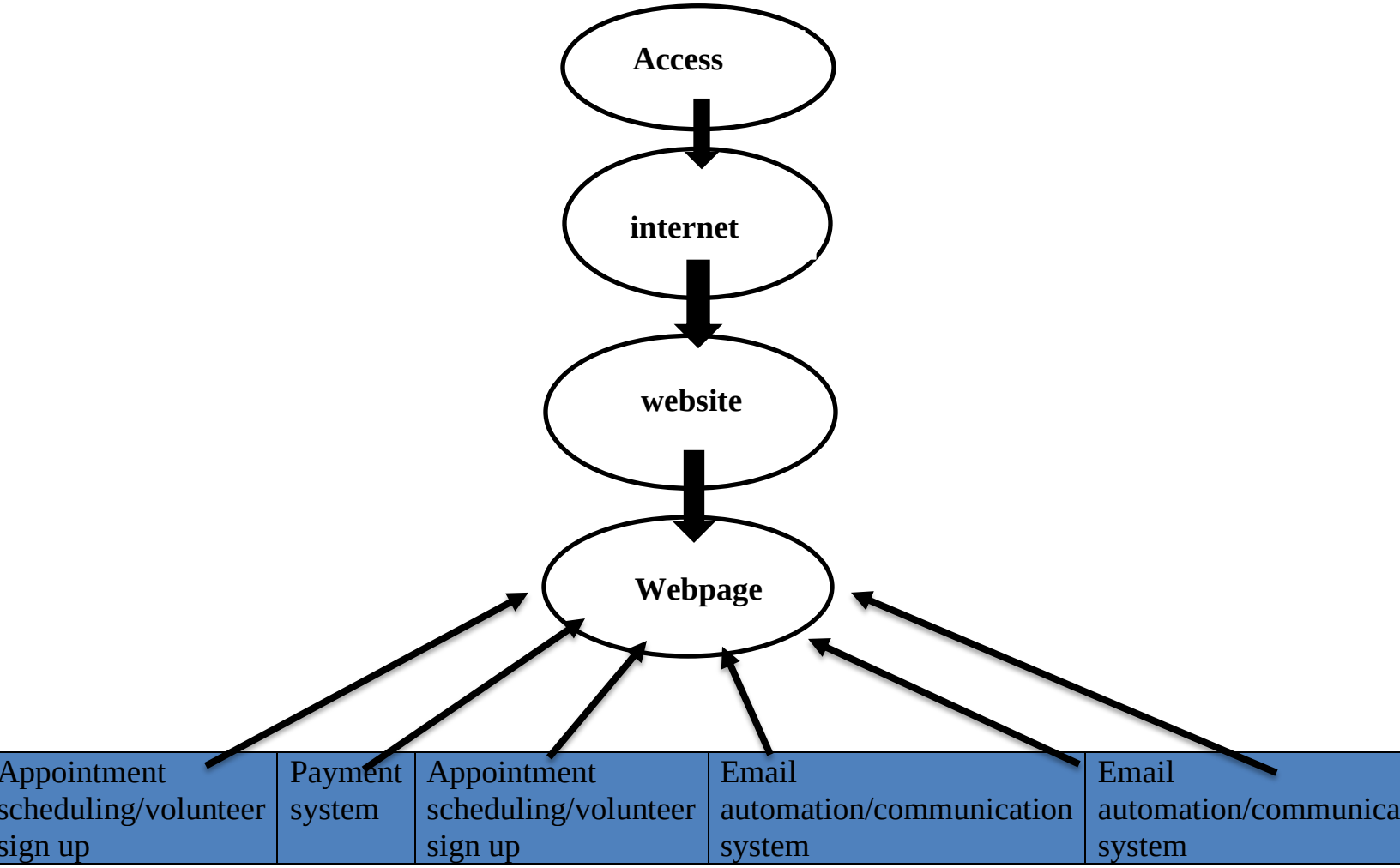
The platform will continue using **Airtable** as its central database, maintaining structured records for pet medical histories, volunteer schedules, and appointment tracking. **Plan Hero** will manage both appointment scheduling and volunteer sign-ups, eliminating the need for separate systems and ensuring a smooth scheduling process for pop-up clinics and staff coordination. **Stripe** will be used to manage payment processing, allowing for secure and efficient transactions related to donations and service fees.

A key enhancement to the system is the integration of **Tidio** as a knowledge-based chatbot that will provide automated responses to frequently asked questions while also processing customer support requests. Unlike traditional chatbots, Tidio will be trained with Pets4Life Louisville-specific information, ensuring that users receive accurate, detailed responses tailored to the organization's services. In cases where Tidio is unable to resolve an inquiry, it will generate a support ticket and forward the request to the appropriate team member for further assistance. This ensures a seamless transition from automated assistance to human intervention when needed. Additionally, Tidio will automate information routing, categorizing and structuring customer concerns before escalation, reducing response time and improving service efficiency.

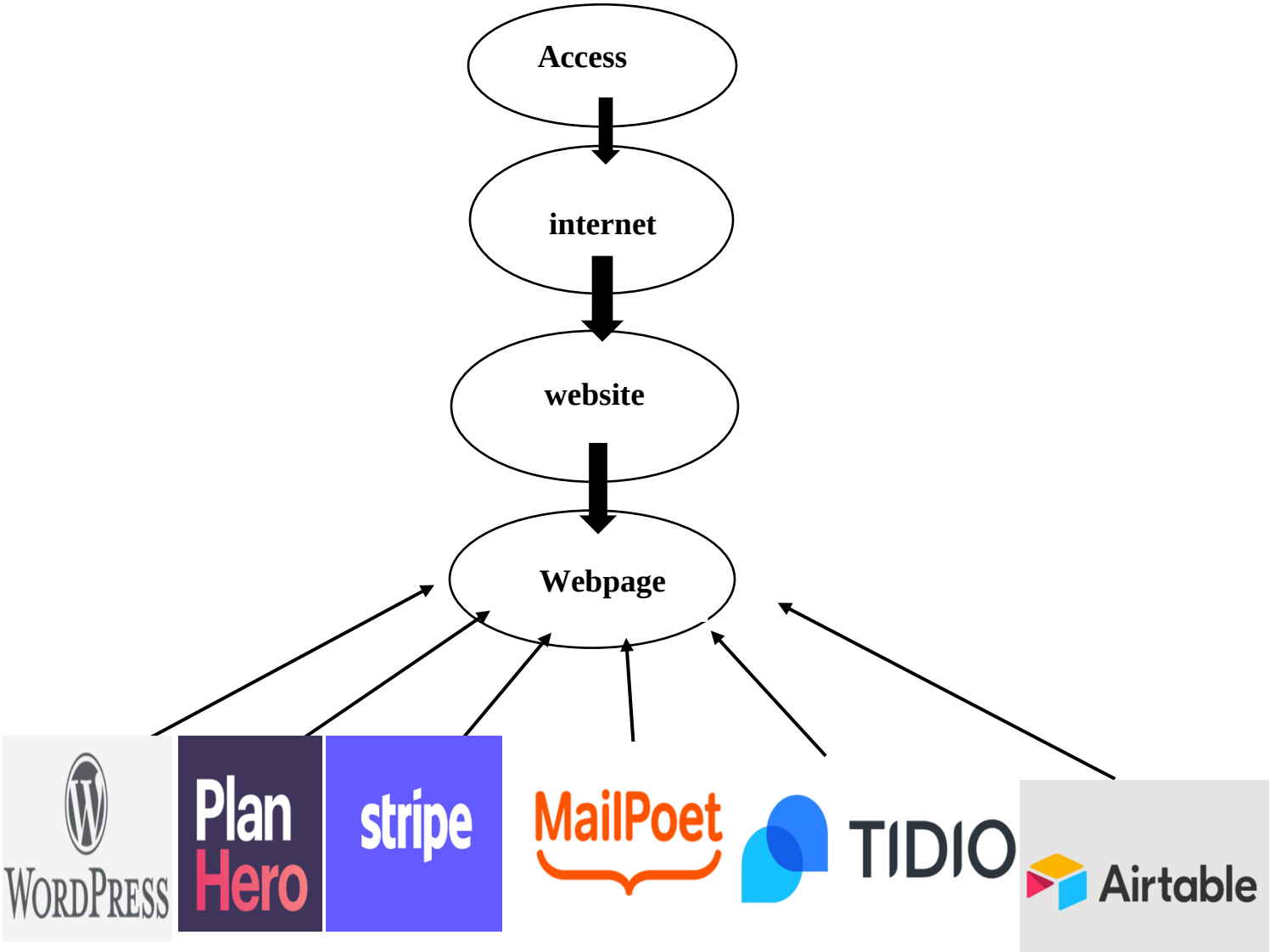
To enhance communication with donors, volunteers, and pet owners, **MailPoet** will be used for email automation, providing event reminders, fundraising updates, and important service notifications. Given that many users will be accessing the platform through mobile devices and in areas with inconsistent internet access, the system is designed to be lightweight and fully mobile-optimized. Security remains a top priority, with SSL encryption, access control measures, and compliance with data privacy regulations ensuring that sensitive information is protected.
The platform is designed with future scalability in mind, allowing for easy expansion to include advanced reporting tools, predictive analytics, and potential integrations with external veterinary service providers. By leveraging these modern

digital solutions, Pets4Life Louisville will be able to optimize its operations while maintaining financial sustainability and user accessibility.

8.1 Initial review



8.2 realistic view



9. Risk Analysis

While the new system offers significant improvements in efficiency and automation, it also introduces several risks that must be addressed to ensure long-term stability and effectiveness. One of the most critical risks is data security, given that the system will store sensitive information such as pet medical records, volunteer details, and financial transactions. Unauthorized access or data breaches could lead to serious privacy violations, financial losses, and damage to the organization's reputation. To mitigate this risk, the platform will employ data encryption, multi-factor authentication, and controlled user access to protect confidential information. Regular security audits will also be conducted to identify and address potential vulnerabilities.

Another major concern is system reliability and uptime, particularly during high-demand periods such as pop-up clinics and major fundraising campaigns. If the system experiences downtime or slow performance during these critical moments, it could disrupt appointment scheduling, donation processing, and customer interactions. To minimize this risk, the platform will be supported by server redundancy, automated backups, and real-time performance monitoring, ensuring that any issues are detected and resolved before they impact operations.

User adoption presents another challenge, as the platform will be used by volunteers, pet owners, and administrative staff with varying levels of technical proficiency. If users find the system difficult to navigate, they may revert to manual processes, reducing the overall efficiency of operations. To facilitate a smooth transition, the system will include an intuitive user interface, multilingual support, and training materials, ensuring that all users can easily access and utilize the available features.

Financial sustainability is also a key risk factor, as Pets4Life Louisville relies on donor contributions to maintain and expand its services. If the new system fails to engage donors effectively, the organization may struggle to fund ongoing maintenance and future improvements. To address this, **automated donor outreach, personalized fundraising campaigns, and transparent financial**

tracking will be implemented to encourage sustained contributions and community involvement.

High-Risk Cases

• **Data Breaches and Unauthorized Access**: Storing confidential user and pet health information makes the system a target for security threats. To prevent unauthorized access, the system will implement encryption, access control restrictions, and multi-factor authentication to safeguard sensitive data.

• **Service Downtime During Critical Events:** If the platform becomes unresponsive during high-traffic events such as pop-up clinics or major fundraisers, essential services could be disrupted. Real-time performance monitoring, load balancing, and failover mechanisms will be in place to ensure continuous operation.

• **Incorrect Ticket Escalation or Delayed Customer Service Responses:** If Tidio fails to escalate support requests properly, customer concerns may go unresolved, impacting service quality. The chatbot will undergo regular training and human oversight, with an emergency escalation feature to ensure no critical issues are ignored.

• **Failed Payment Processing and Donation Errors:** Any issues with Stripe could result in failed transactions, donor frustration, and financial discrepancies. Secure API connections, transaction logging, and fraud detection mechanisms will be used to maintain payment reliability.

• **Volunteer Scheduling Conflicts:** Poor shift management could leave clinics understaffed, affecting service efficiency. Plan Hero’s automated scheduling, real-time notifications, and availability tracking will help ensure proper coordination.

• **Low User Adoption Due to Technical Barriers:** If users struggle with the platform, they may opt for traditional manual methods, reducing its effectiveness. Providing training materials, chatbot assistance, and a simplified onboarding process will help users feel more comfortable with the transition.

Low-Risk Cases

• **Minor Chatbot Errors:** While Tidio is designed to provide accurate responses, occasional misinterpretations may occur. These will be addressed through ongoing chatbot training and improvements.

• **Performance Differences Across Devices:** Some users may experience slightly slower load times depending on their device and network speed. However, mobile optimization will ensure a consistent and responsive experience for most users.

• **Delayed Rollout of Additional Features:** Some advanced functionalities may take longer to develop, but the core system will remain operational, and updates will be introduced progressively.

• **Occasional Volunteer No-Shows:** Unexpected cancellations may cause minor disruptions, but Plan Hero's flexible scheduling system and backup volunteer assignments will help mitigate the impact.

10. System Requirements

Functional Requirements

Client and Pet Onboarding

- **ID: SR001** - Enable new clients to register, creating profiles for themselves and their pets.
- **ID: SR002** - Allow clients to upload veterinary records and other relevant pet information.
- **ID: SR003** - Enable profile updates by clients, including contact information and pet details.
- **ID: SR004** - Provide a system for password recovery and account security management.

Donation Management

- **ID: SR005** - Process and record one-time and recurring donations.
- **ID: SR006** - Automatically generate and send digital receipts for donations.

- **ID: SR007** - Allow donors to select specific programs or funds they wish to support.
- **ID: SR008** - Provide administrative tools for tracking and analyzing donation patterns.

Volunteer Management

- **ID: SR009** - Enable volunteer registration with detailed profile creation.
- **ID: SR010** - Calendar integration for scheduling volunteer activities.
- **ID: SR011** - Automated notifications for volunteer event reminders.
- **ID: SR012** - System for tracking volunteer hours and contributions.

Event Management

- **ID: SR013** - Allow staff to create and manage public events, including details such as date, time, location, and participant limits.
- **ID: SR014** - Online registration and payment for events.
- **ID: SR015** - Automated confirmation and reminder emails to event participants.
- **ID: SR016** - Features to promote events on social media and other platforms.

Pet Adoption Process

- **ID: SR017** - Display detailed profiles of sponsorable pets, including photos and health information.
- **ID: SR018** - Administrative interface for managing adoption applications and status updates.

- **ID: SR019** - Automated notifications to applicants about their application status.

Customer Support

- **ID: SR020** - Chatbot support for immediate assistance.
- **ID: SR021** - Ticketing system for handling and tracking customer inquiries.
- **ID: SR022** - FAQ section for common queries and concerns.
- **ID: SR023** - Feedback system for customers to rate services and provide suggestions.

Content Management

- **ID: SR024** - Tools for staff to post updates, blogs, and pet care information.
- **ID: SR025** - Photo and video gallery management for event and pet highlights.
- **ID: SR026** - Secure management of user-generated content, including reviews and comments.
- **ID: SR027** - Editorial tools for content revision and approval.

Administrative and Reporting Tools

- **ID: SR028** - Comprehensive dashboards for monitoring website traffic and user activity.
- **ID: SR029** - Detailed reporting tools for financial management, including donations and event revenues.

11. Use Case Scenarios

Use Case 1: Client Registration

- **Primary Actor**: Pet Owner
- **Brief Description**: Allows new clients to create profiles for themselves and their pets.
- **Risk Level:** Low

Use Case 2: Upload Veterinary Records

- **Primary Actor**: Pet Owner
- **Brief Description**: Enables clients to upload veterinary records and other relevant pet information.
- **Risk Level:** Medium

Use Case 3: Profile Management

- **Primary Actor**: Pet Owner
- **Brief Description**: Allows clients to update their profiles, including contact information and pet details.
- **Risk Level:** Low

Use Case 4: Password Recovery

- **Primary Actor**: Pet Owner

- **Brief Description**: Provides a system for password recovery and account security management.
- **Risk Level:** High

Use Case 5: Process Donations

- **Primary Actor**: Donor
- **Brief Description**: Allows donors to make one-time and recurring donations through the system.
- **Risk Level:** High

Use Case 6: Generate Donation Receipts

- **Primary Actor**: System
- **Brief Description**: Automatically generates and sends digital receipts for donations.
- **Risk Level:** Low

Use Case 7: Donation Allocation

- **Primary Actor**: Donor
- **Brief Description**: Enables donors to select specific programs or funds they wish to support.
- **Risk Level:** Low

Use Case 8: Donation Analytics

- **Primary Actor**: Fundraising Manager
- **Brief Description**: Provides administrative tools for tracking and analyzing donation patterns.
- **Risk Level:** Medium

Volunteer Management

Use Case 9: Volunteer Registration

- **Primary Actor**: Volunteer
- **Brief Description**: Enables volunteers to register with detailed profile creation.
- **Risk Level:** Low

Use Case 10: Volunteer Scheduling

- **Primary Actor**: Volunteer Coordinator
- **Brief Description**: Integrates a calendar system for scheduling volunteer activities.
- **Risk Level:** Low

Use Case 11: Volunteer Notifications

- **Primary Actor**: System
- **Brief Description**: Sends automated notifications for volunteer event reminders.
- **Risk Level:** Low

Use Case 12: Volunteer Tracking

- **Primary Actor**: Volunteer Coordinator
- **Brief Description**: Tracks volunteer hours and contributions for record-keeping.
- **Risk Level:** Low

Event Management

Use Case 13: Event Creation

- **Primary Actor**: Staff Member
- **Brief Description**: Allows staff to create and manage public events with relevant details.
- **Risk Level:** Medium

Use Case 14: Event Registration & Payment

- **Primary Actor**: Pet Owner / Donor

- **Brief Description**: Enables users to register and make payments for events online.
- **Risk Level:** High

Use Case 15: Event Notifications

- **Primary Actor**: System
- **Brief Description**: Sends automated confirmation and reminder emails to event participants.
- **Risk Level:** Low

Use Case 16: Event Promotion

- **Primary Actor**: Marketing Staff
- **Brief Description**: Provides tools to promote events via social media and other platforms.
- **Risk Level:** Low

Pet Adoption Process

Use Case 17: Display Pet Profiles

- **Primary Actor**: Public User
- **Brief Description**: Displays detailed profiles of available pets for adoption.

- **Risk Level:** Low

Use Case 018: Adoption Management

- **Primary Actor**: Shelter Staff
- **Brief Description**: Enables staff to manage adoption applications and update status.
- **Risk Level:** Medium

Use Case 19: Adoption Notifications

- **Primary Actor**: System
- **Brief Description**: Sends automated notifications to applicants about their application status.
- **Risk Level:** Low

Customer Support

Use Case 20: Chatbot Assistance

- **Primary Actor**: Pet Owner
- **Brief Description**: Provides immediate customer support via chatbot.
- **Risk Level:** Medium

Use Case 21: Ticketing System

- **Primary Actor**: Customer Support Staff
- **Brief Description**: Implements a ticketing system to track customer inquiries.
- **Risk Level:** Medium

Use Case 22: FAQ Section

- **Primary Actor**: Public User
- **Brief Description**: Provides a knowledge base for common queries and concerns.
- **Risk Level:** Low

Use Case 23: Customer Feedback

- **Primary Actor**: Pet Owner
- **Brief Description**: Allows customers to submit service feedback and ratings.
- **Risk Level:** Low

Content Management

Use Case 24: Publish Content

- **Primary Actor**: Staff Member
- **Brief Description**: Allows staff to post updates, blogs, and pet care information.
- **Risk Level:** Low

Use Case 25: Media Management

- **Primary Actor**: Staff Member
- **Brief Description**: Manages photo and video galleries for events and pet highlights.
- **Risk Level:** Low

Use Case 26: User-Generated Content

- **Primary Actor**: System Administrator
- **Brief Description**: Manages user-generated content, including reviews and comments.
- **Risk Level:** Medium

Use Case 27: Editorial Control

- **Primary Actor**: Content Manager
- **Brief Description**: Provides editorial tools for content revision and approval.
- **Risk Level:** Low

Administrative and Reporting Tools

Use Case 28: Website Analytics Dashboard

- **Primary Actor**: Administrator
- **Brief Description**: Monitors website traffic and user activity.
- **Risk Level:** Medium

Use Case 29: Financial Reports

- **Primary Actor**: Administrator
- **Brief Description**: Generates reports for financial management, including donations and event revenues.
- **Risk Level:** High

12. Trace Matrix

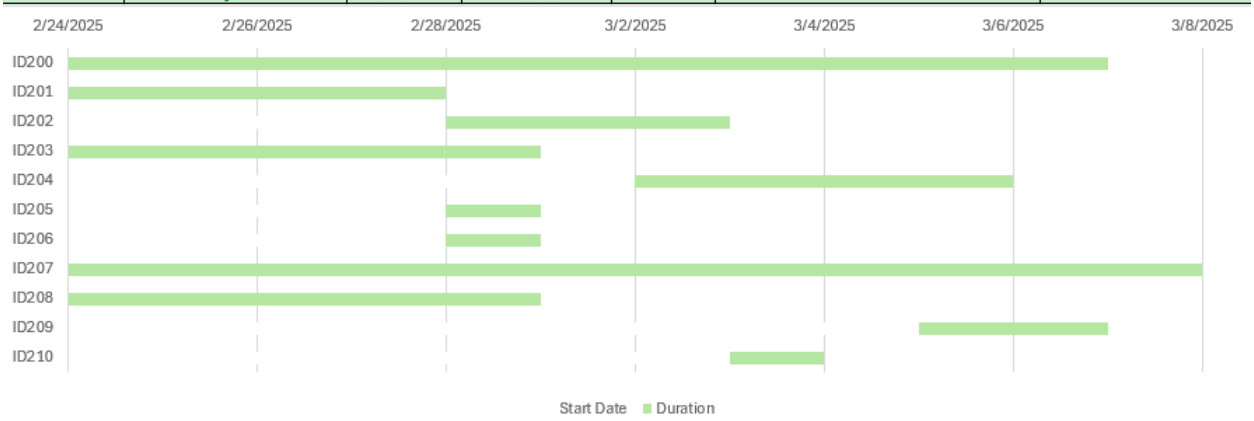
[Link to excel sheet]

	Use Case 1: Client Registration	Use Case 2: Upload Veterinary Records	Use Case 3: Profile Management	Use Case 4: Password Recovery	Use Case 5: Process Donations	Use Case 6: Generate Donation Receipts	Use Case 7: Donation Allocation	Use Case 8: Donation Analytics	Use Case 9: Volunteer Registration	Use Case 10: Volunteer Scheduling	Use Case 11: Volunteer Notifications	Use Case 12: Volunteer Tracking	Use Case 13: Event Creation	Use Case 14: Event Registration & Payment	Use Case 15: Event Notifications	Use Case 16: Event Promotion	Use Case 17: Display Pet Profiles	Use Case 18: Adoption Management	Use Case 19: Adoption Notifications	Use Case 20: Chatbot Assistance	Use Case 21: Ticketing System	Use Case 22: FAQ Section	Use Case 23: Customer Feedback	Use Case 24: Publish Content	Use Case 25: Media Management	Use Case 26: User-Generated Content	Use Case 27: Editorial Council	Use Case 28: Website Analytics Dashboard	Use Case 29: Financial Reports	TOTAL
ID: SR001 - Enable new clients to register, creating profiles for themselves and their pets	✓																													1/29
ID: SR002 - Allow clients to upload veterinary records and other relevant pet information		✓																												1/29
ID: SR003 - Enable profile updates by clients, including contact information and pet details			✓																											1/29
ID: SR004 - Provide a system for password recovery and account security management.				✓																										1/29
ID: SR005 - Process and record one-time and recurring donations					✓																									1/29
ID: SR006 - Automatically generate and send digital receipts for donations						✓																								1/29
ID: SR007 - Allow donors to select specific programs or funds they wish to support							✓																							1/29
ID: SR008 - Provide administrative tools for tracking and analyzing donation patterns								✓																						1/29
ID: SR009 - Enable volunteer registration with detailed profile creation									✓																					1/29
ID: SR010 - Calendar integration for scheduling volunteer activities										✓																				1/29
ID: SR011 - Automated notifications for volunteer event reminders											✓																			1/29
ID: SR012 - System for tracking volunteer hours and contributions												✓																		1/29
ID: SR013 - Allow staff to create and manage public events, including details such as date, time, location, and participant limits													✓																	1/29
ID: SR014 - Online registration and payment for events														✓																1/29
ID: SR015 - Automated confirmation and reminder emails to event participant															✓															1/29
ID: SR016 - Features to promote events on social media and other platforms																✓														1/29
ID: SR017 - Display detailed profiles of sponsorable pets, including photos and health information																	✓													1/29
ID: SR018 - Administrative interface for managing adoption applications and status updates																		✓												1/29
ID: SR019 - Automated notifications to applicants about their application status																			✓											1/29
ID: SR020 - Chatbot support for immediate assistance																				✓										1/29
ID: SR021 - Ticketing system for handling and tracking customer inquiries																					✓									1/29
ID: SR022 - FAQ section for common queries and concerns																						✓								1/29
ID: SR023 - Feedback system for customers to rate services and provide suggestions																							✓							1/29
ID: SR024 - Tools for staff to post updates, blogs, and pet care information																								✓						1/29
ID: SR025 - Photo and video gallery management for event and pet highlights																									✓					1/29
ID: SR026 - Secure management of user-generated content, including reviews and comments																										✓				1/29
ID: SR027 - Editorial tools for content revision and approval																											✓			1/29
ID: SR028 - Comprehensive dashboards for monitoring website traffic and user activity																												✓		1/29
ID: SR029 - Detailed reporting tools for financial management, including donations and event revenues																													✓	1/29
TOTAL	1/29	1/29	1/29	1/29	1/29	1/29	1/29	1/29	1/29	1/29	1/29	1/29	1/29	1/29	1/29	1/29	1/29	1/29	1/29	1/29	1/29	1/29	1/29	1/29	1/29	1/29	1/29	1/29	1/29	

13. Gantt Chart

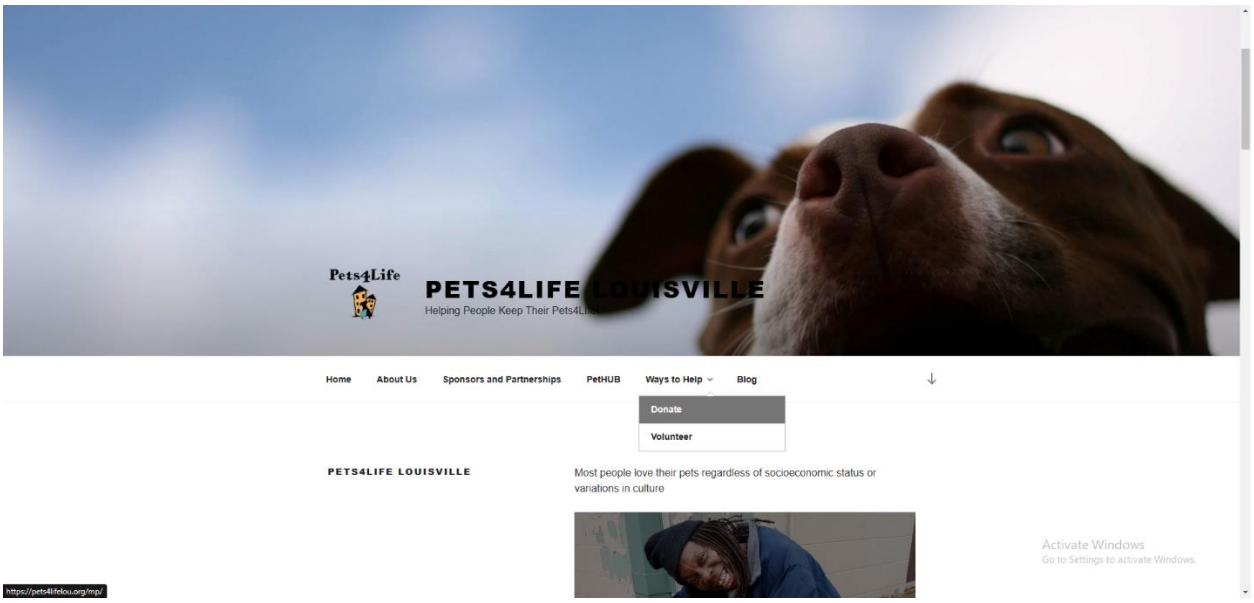
[Link to Excel sheet]

Task ID	Task	Start Date	End date	Duration	Task Responsibilty	Dependency
ID200	I2 Document Complile	2/24/2025	3/7/2025	11	Isaiah Martinez	
ID201	Vision Document	2/24/2025	2/28/2025	4	Isaiah Martinez	
ID202	Feasibility Analysis	2/28/2025	3/3/2025	3	Greiler Abrahantes and Eric Cato	ID201
ID203	System Requirements	2/24/2025	3/1/2025	5	Greiler Abrahantes and Eric Cato	ID201
ID204	Uses Cases	3/2/2025	3/6/2025	4	Greiler Abrahantes and Eric Cato	ID203
ID205	Initial Architecture	2/28/2025	3/1/2025	1	Ozine Byamungu	
ID206	Risk Analysis	2/28/2025	3/1/2025	1	Ozine Byamungu	ID204
ID207	Gantt Chart	2/24/2025	3/8/2025	12	Kyah Owens	
ID208	Inception Prototype	2/24/2025	3/1/2025	5	Kyah Owens and Isaiah Martinez	
ID209	Trace Matrix	3/5/2025	3/7/2025	2	Kyah Owens	ID204, ID203
ID210	NPV Analysis	3/3/2025	3/4/2025	1	Eric Cato	

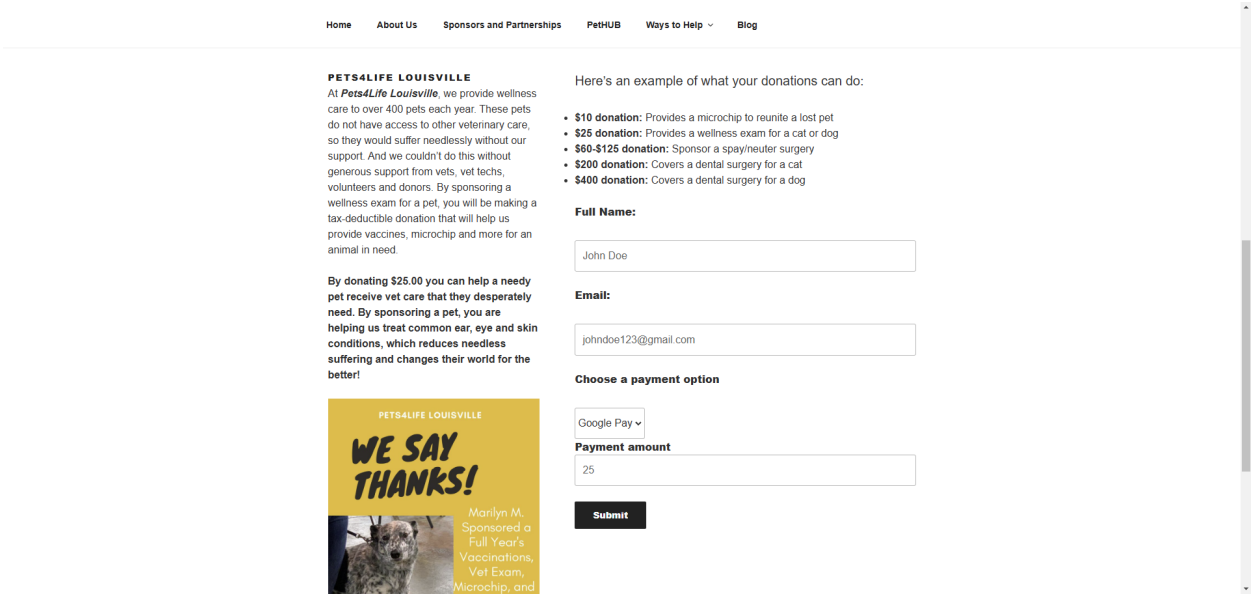


14. Inception Phase

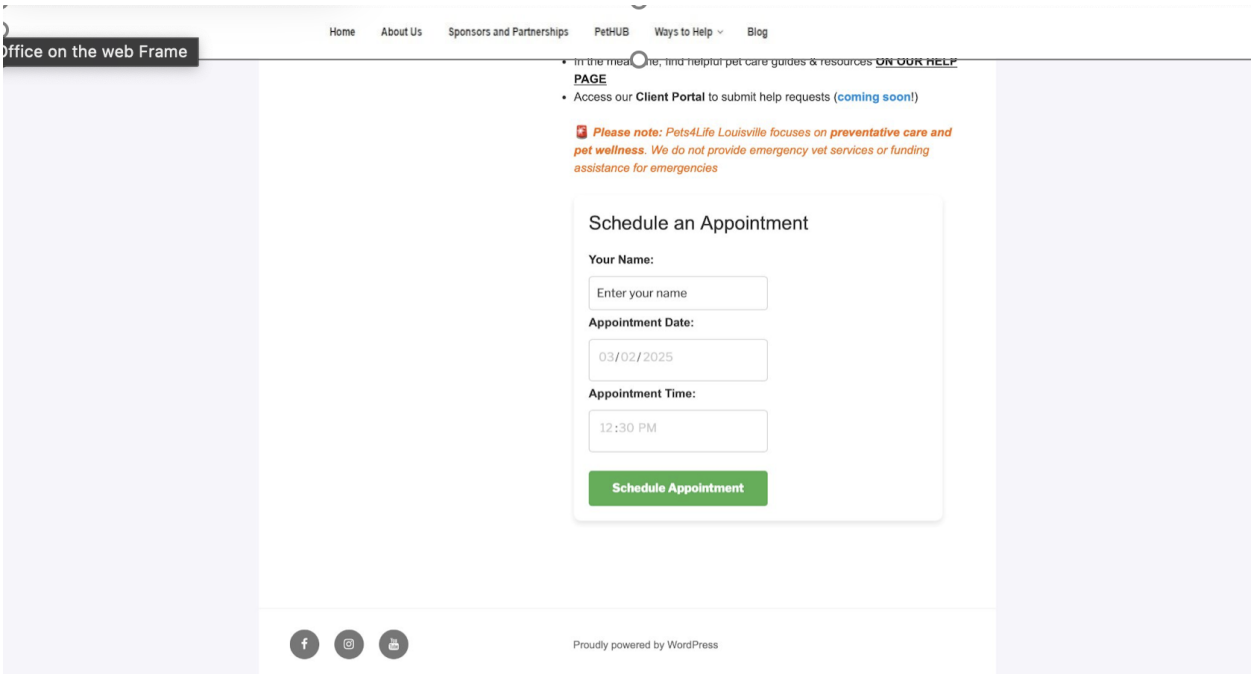
Landing page



Donation page



Appointment scheduling page



15. Appendix A

NPV Analysis:

The purpose of conducting the NPV analysis for Pets4Life was to evaluate the financial feasibility of the project by assessing projected revenues and expenses, discounted to present value. This analysis helps determine whether the organization can sustain itself financially while achieving its mission. The NPV calculations considered three different scenarios, each reflecting varying levels of revenue growth and cost reduction to assess the potential financial outcomes.

The first NPV analysis assumed a **10% annual revenue increase and a 20% expense reduction**, resulting in:

- **Total NPV:** $239,440.39
- **ROI:** 21%
- **Break-Even Point (BEP):** Year 1

The second NPV analysis accounted for a **15% annual revenue increase and a 25% expense reduction**, which improved financial outcomes:

- **Total NPV:** $317,074.02

- **ROI:** 88.3%
- **Break-Even Point (BEP):** Year 1

  The third and most favorable scenario projected a **20% annual revenue increase and a 30% expense reduction**, leading to:

- **Total NPV:** $400,833.57
- **ROI:** 180.6%
- **Break-Even Point (BEP):** Year 1

  Across all scenarios, the analysis confirmed that Pets4Life achieves break-even within the first year, demonstrating strong financial viability. Additionally, higher revenue growth and cost reductions significantly improve long-term sustainability, making the project a viable investment for future expansion.

Year	Revenue	Expenses	Net Income	Yearly NPV (10% Discount Rate)
1	$87,835	$58,864.80	$28,970.20	**$26,336.55**
2	$96,618.50	$47,091.84	$49,526.66	**$41,056.36**
3	$106,280.35	$37,673.47	$68,606.88	**$50,991.89**
4	$116,908.39	$30,138.79	$86,769.61	**$57,469.85**
5	$128,599.23	$24,111.02	$104,488.21	**$63,585.74**

Revenue Increase: **10%** annually
Expenses Decrease: **20%** annually

Total NPV: **$239,440.39**
Total Expenses: **$197,879.91**
ROI: **21.0%**
BEP: **Year 1**

Year	Revenue	Expenses	Net Income	Yearly NPV (10% Discount Rate)
1	$91,827.50	$55,185.75	$36,641.75	$33,310.68
2	$105,601.63	$41,389.31	$64,212.32	$52,955.74
3	$121,441.87	$31,041.98	$90,399.89	$66,962.04
4	$139,658.15	$23,281.49	$116,376.66	$76,784.47
5	$160,606.87	$17,461.11	$143,145.76	$87,061.09

Revenue Increase: **15%** annually
Expenses Decrease **25%** annually

Total NPV: **$317,074.02**
Total Expenses: **$168,359.64**
ROI: **88.3%**
BEP: **Year 1**

Year	Revenue	Expenses	Net Income	Yearly NPV (10% Discount Rate)
1	$95,820	$51,506.70	$44,313.30	$40,284.82
2	$114,984	$36,054.69	$78,929.31	$65,212.11
3	$137,980.80	$25,238.28	$112,742.52	$84,219.89
4	$165,576.96	$17,666.80	$147,910.16	$97,617.91
5	$198,692.35	$12,366.76	$186,325.59	$113,498.84

Revenue Increase: **20%** annually
Expenses Decrease **30%** annually

Total NPV: **$400,833.57**
Total Expenses: **$142,833.23**
ROI: **180.6%**
BEP: **Year 1**

	SUMMARY		
	10% Annual Revenue Increase & 20% Annual Cost Reduction	15% Annual Revenue Increase & 25% Annual Cost Reduction	20% Annual Revenue Increase & 30% Annual Cost Reduction
TOTAL NPV	$ 239,440.39	$317,074.02	$ 400,833.57
ROI	21%	88.3%	180.6%
BEP	Year1	Year 1	Year 1